

In Kyu Lee

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Education

- Ph.D., University of California San Diego, La Jolla, CA** Sep 2024 – Present
- Major: Mechanical Engineering
 - Advisor: Claire Acevedo, PhD
- M.S., University of Kansas, Lawrence, KS** Jul 2022
- Major: Bioengineering, *with Honors*
 - Advisor: Jiwoong Choi, PhD
- B.S., University of Minnesota–Twin Cities, Minneapolis, MN** May 2020
- Major: Aerospace Engineering and Mechanics | Minor: Computer Science
 - Advisor: Hyun Soo Park, PhD

Experience

- Graduate Research Assistant — F2 Lab, UC San Diego** Aug 2024 – Present
- Developed a segmentation and quantification pipeline for high-resolution microCT
 - Analyzed how chronic lymphocytic leukemia (CLL) affects bone and the marrow microenvironment using synchrotron microCT
- Research Scientist — Medipixel** Aug 2022 – Aug 2024
- Developed a frame selection module for coronary angiography based on image quality assessment
 - Developed a local refinement module for lesion and bifurcation regions
 - Led the team entry to the MICCAI 2023 ARCADE Challenge (1st place, stenosis segmentation; 2nd place, coronary artery segmentation)
 - Developed a video-based coronary complication detection model
 - Automated a computational fluid dynamics (CFD) pipeline using ANSYS Fluent (PyFluent)
- Graduate Research Assistant — University of Kansas Medical Center** Aug 2020 – Jul 2022
- Built a quantitative CT (QCT) analysis pipeline for the lung, including:
 - Image registration
 - Machine learning and statistical analysis
 - 1D airway CFD simulation
- Research Assistant — Seoul National University Hospital** Summer 2019, Summer 2020
- Advisor: Chang Hyun Lee, MD
 - Participated in the Korea Environmental Industry & Technology Institute (KEITI) project “CT Image Matching Technology Development for Structural and Functional Assessment of Environmentally Induced Lung Disease,” funded by the Korean Ministry of Environment
- Undergraduate Research Assistant — Computer Vision, University of Minnesota** Jan 2019 – Mar 2020
- Processed the Human Multiview Behavior Imaging (HUMBI) dataset
 - Reconstructed 3D human body meshes from multiview 2D camera images

Teaching

Graduate Teaching Assistant — UC San Diego

- MAE 290A: Numerical Linear Algebra Fall 2026
- BENG/MAE 209: Continuum Mechanics Applied to Medicine & Biology Spring 2026

Undergraduate Teaching Assistant — University of Minnesota

Spring 2019

- AEM 3031: Deformable Body Mechanics

Peer Tutor — SMART Learning Commons, University of Minnesota

Sep 2018 – May 2020

- Tutored undergraduate science and calculus courses

Publications

* Equal contribution

Thesis

Lee IK. Extended Quantitative Computed Tomography Analysis of Lung Structure and Function. M.S. thesis, University of Kansas, **2022**.

Peer-Reviewed Papers

Lee IK, Obata Y, Pmictier A, Williams J, Kikani B, Jensen C, Sborov D, Stephens D, Acevedo C. Disruption of Marrow Microenvironments in Chronic Lymphocytic Leukemia by High-Resolution Synchrotron Micro-Computed Tomography. *Communications Medicine*, **2026** (accepted).

Lee IK*, Seo S*, Min J. Anatomy-Grounded Synthetic Coronary Angiography for Geometry-Informed Multi-View Matching. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, **2026**.

Seo S, **Lee IK**, Kim H-W, Min J, Jung C-H. Diffusion-Based User-Guided Data Augmentation for Coronary Stenosis Detection. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, **2025**.

Lee IK, Ku J, Choi Y. Gray's Anatomy for Segment Anything Model: Optimizing Grayscale Medical Images for Fast and Lightweight Segmentation. *CVPR 2024 Challenge: Segment Anything in Medical Images on Laptop (MedSAM on Laptop)*, LNCS vol. 15458, **2024**.

Yu Z*, Yoon JS*, **Lee IK**, Venkatesh P, Park J, Yu J, Park HS. HUMBI: A Large Multiview Dataset of Human Body Expressions. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, virtual, June 16–18, **2020**.

Preprints

Lee IK, Shin J, Lee Y-H, Ku JH, Kim H-W. SSASS: Semi-Supervised Approach for Stenosis Segmentation. *Automatic Region-based Coronary Artery Disease diagnostics using x-ray angiography imagEs challenge (ARCADE)*, **MICCAI Challenge 2023**. [arXiv:2311.10281](https://arxiv.org/abs/2311.10281)

Ku JH, Lee Y-H, Shin J, **Lee IK**, Kim H-W. MPSEg: Multi-Phase Strategy for Coronary Artery Segmentation. *Automatic Region-based Coronary Artery Disease diagnostics using x-ray angiography imagEs challenge (ARCADE)*, **MICCAI Challenge 2023**. [arXiv:2311.10306](https://arxiv.org/abs/2311.10306)

Conference Abstracts and Presentations

Lee IK, Obata Y, Pmictar A, Williams J, Kikani B, Jensen C, Sborov D, Acevedo C, Stephens D. Disruption of Marrow Microenvironments in Chronic Lymphocytic Leukemia by High-Resolution Synchrotron Micro-Computed Tomography. *Society of Engineering Science (SES) Annual Technical Meeting*, Atlanta, GA, Oct 12–15, **2025 (oral)**.

Lee IK, Choi J, Castro M, Kim T, Kang HR, Lee CH. Quantitative CT and CFD Explain Bronchodilator-Induced Regional Ventilation Improvement in Asthma. *International Collaborative Asthma Network (ICAN)*, San Francisco, CA, May 12–13, **2022 (oral)**.

Lee IK, Choi J, Park EK, Tutkun E, Hoffman EA, Castro M, Lee CH. Toward Extended Quantitative CT Imaging Application with Deep Transfer Learning. *American Thoracic Society (ATS) International Conference*, San Diego, CA (virtual), May 14–19, **2021**. Am J Respir Crit Care Med 2021;203:A4591 (poster).

Honors & Awards

American Society of Hematology (ASH) Abstract Achievement Award	Oct 2025
MICCAI 2023 ARCADE Challenge • Stenosis segmentation: 1st place • Coronary artery segmentation: 2nd place	Oct 2023
2023 Social Issues Data & AI Analysis Competition • 2nd place (\$1,600), awarded by the President of the National Research Council	Oct 2023
Outstanding Student Researcher Award, University of Kansas	May 2023
WSI Lung Cancer Segmentation Competition, Ajou University Medical Center • 2nd place (\$2,400)	Dec 2022
Burn Diagnosis AI Challenge, Ministry of Science and ICT, Korea • 1st place (\$2,400)	Nov 2022
International Collaborative Asthma Network (ICAN) Travel Award	Mar 2022
Chester Gaskell Aeronautical Engineering Scholarship • Merit-based scholarship for top undergraduates with outstanding academic records	Fall 2019
Undergraduate Research Opportunities Program (UROP) Award	May 2019
Dean's List, University of Minnesota • Fall 2014, Spring 2015, Fall 2017, Spring 2018, Fall 2018, Spring 2019	

Activities

Kaggle Competitions Expert	
Plaque of Appreciation, President of the Korean American Society of Greater Kansas City	Jul 2022

Patents & Certifications

Patents

- Method and apparatus for matching medical images considering phase of respiration signal
- Method and electronic device for labeling medical image
- Prediction model training method, FFR data generating method and electronic device performing the same

Certifications

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| • NVIDIA: Fundamentals of Accelerated Computing with CUDA Python | Jun 2021 |
| • Udacity: AI for Healthcare Nanodegree | Sep 2020 |
| • College Reading & Learning Association (CRLA): Certified Level III Master Tutor | May 2019 |

Skills

Languages

- Korean (native), English (fluent), Chinese (fluent)

Programming

- Python, MATLAB, Fortran